

Stardog Advanced Designer Training Exercise Guide

This exercise guide is designed to provide examples on how to handle complex data modeling and mapping in Stardog Designer. The exercises will cover examples on the following topics: mapping a single flat resource, creating a taxonomy/hierarchy and modeling and mapping a many to many relationships.

Please note: *The Cloud applications are constantly changing and though we try to keep this guide up to date, sometimes the screenshots and steps will vary. In this case, email training@stardog.com for assistance.*

Training Environment

This course will use Stardog Cloud, Explorer and Designer. It is recommended that users use a Stardog Free Account for these exercises.

Get-started: <https://cloud.stardog.com/get-started>

Stardog Cloud: <http://cloud.stardog.com>

Download these files locally:

- [Students.csv](#)
- [geography taxonomy.csv](#)
- [Courses.csv](#)
- [links.csv](#)

Mapping a single .csv resource

This example will show how to map a single .csv resource using the create and map feature in Designer. When you use this feature, Designer suggests the mappings for the file. These suggestions can then be reviewed and edited by the user.

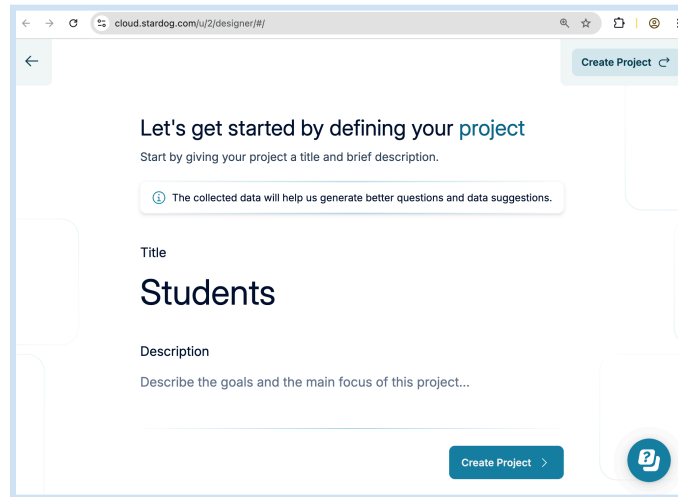
Exercise 1 - Create a New Designer Project

Step 1 - Open Designer

- From <https://cloud.stardog.com/>, click on `Stardog Designer`.

Step 2 - Create a new project

- From the Designer home screen, click on the `New` button in the upper right corner.

A screenshot of a web browser showing the 'Create Project' form in the Stardog Designer. The browser's address bar shows 'cloud.stardog.com/ui/2/designer/#/'. The form has a title field with the text 'Students' and a description field. A 'Create Project' button is at the bottom right. A notification bubble at the top says 'Let's get started by defining your project' and 'Start by giving your project a title and brief description.' A small info bubble says 'The collected data will help us generate better questions and data suggestions.'

- **Title** your project. For example: **Students**
- Optionally enter a **Description**.
- Choose **`Create Project`**

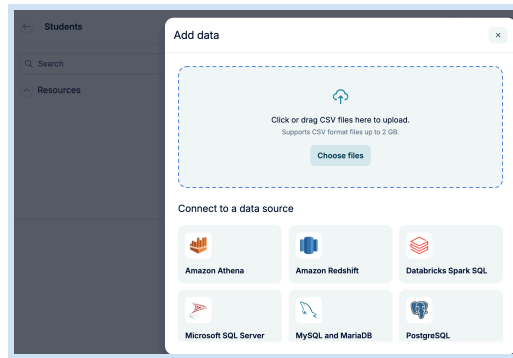
Exercise 2 - Map a .csv Resource

Step 1 - Download the .csv file

- [students.csv](#)

Step 2 - Click the + next to Resources to add a new project resource

- Click on Choose Files and navigate to the **student.csv** file



- Click on Create & Map
- Accept the suggestions with **Accept Changes**
- Once you see the class and the resource on the canvas, click on Mapping

 Data Model
  Mapping

- Click on the **students - students** mapping
- Review the auto-mapping

Exercise 3 - Edit the attributes

Now it's time to review the auto-generated mappings. Verify that the primary key is correct and the label is a good choice of label. It's also important to confirm that each attribute has the appropriate data type to support accurate querying, reasoning, and future analysis.

Step 1 - Edit Attribute Data Types

To edit an attribute, click on the Attribute name.

- Change the data type for **Average Score** to **decimal**
- Change the data type for **Date** to **date**
- Change the data type for **Remote** to **boolean**

Class Detail

Basic

Hierarchy

Relationships

Attributes

Geospatial

Attributes

Attribute Name	Data Type	Rule
		Add
Age	integer	Create
Average score	decimal	Create
Date	date	Create
Gender	string	Create
Remote	boolean	Create
Student ID	integer	Create
Student Name	string	Create

Step 2 - Publish the Project

It is time to check your work by publishing the project and viewing it in Explorer.

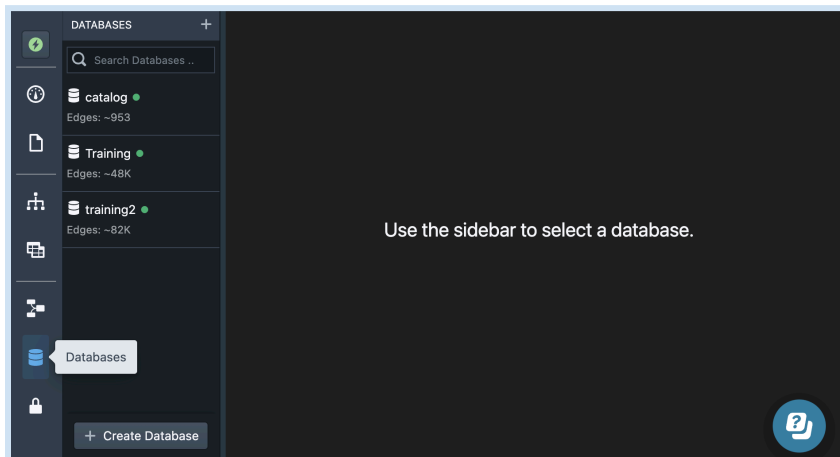
- Click on Publish
- Select or create a database
 - Please note that with a Stardog Free account, you can create up to three user databases, in addition to any system databases that come with your endpoint (such as catalog or querylog). If you have already reached this limit, you can either save your work to one of your existing databases or delete a database you no longer need via Stardog Studio. See the optional step below for guidance.
 - If you create a new database, after entering a new database name, click the + to create it.
- Configure Model - select the **Create new model**
 - Note: The **Download zipped project when publishing** option is a great way to download a .zip file with all the files you need for your project. It is recommended to keep a copy of the latest project files. For this training exercise, the option can be turned off.
- The publishing workflow



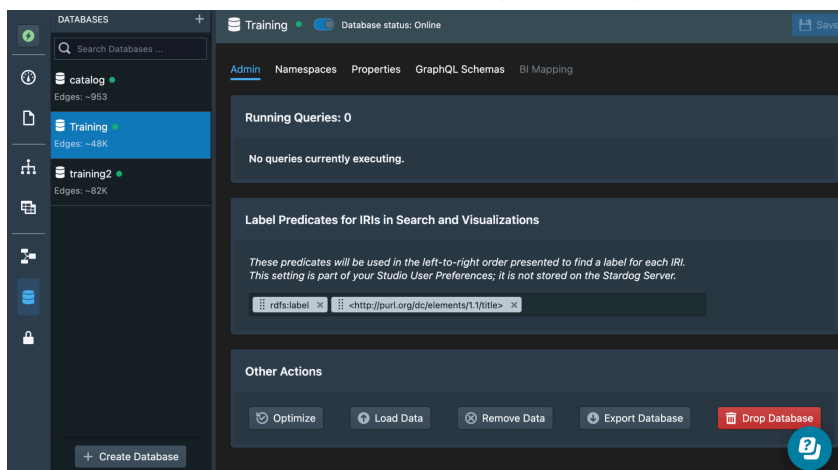
- Accept the defaults and click Publish
 - Open in Explorer. Right-click on the **Student** class and select **expand by** → **instances of** and select any instance. Then right click the instance and choose **see details** to verify that the mapping looks correct.

Optional Step - Delete a database in Stardog Studio

- Go to the Stardog Cloud browser tab and click and open Studio
- Navigate to the Database icon on the left navigation



- Click on the database that you want to delete
- Click on Drop Database and then say "Yes" if you want to remove it.

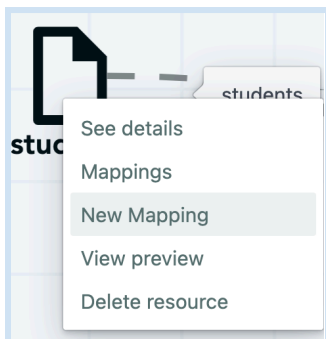


Exercise 4 - Create and Map a New Class

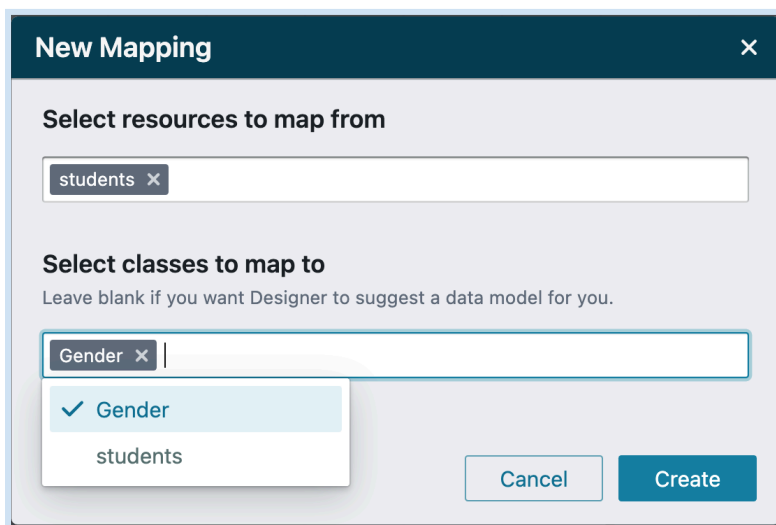
In this exercise you will add and map a new class.

Step 1 - Add Gender as a Class

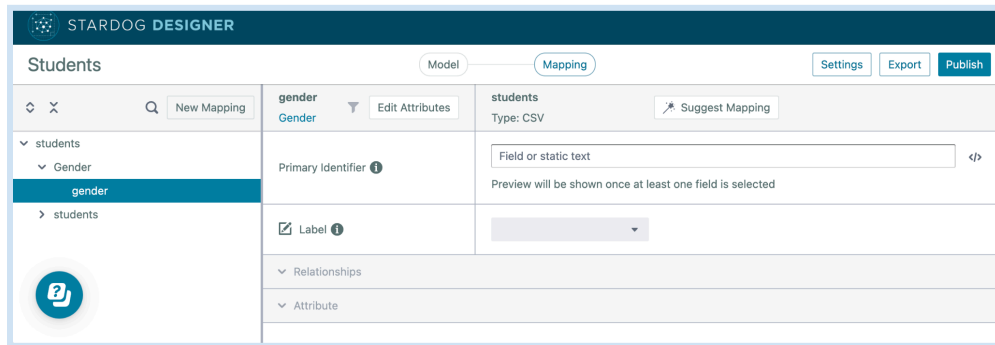
- Click on  to add a new class
- Class Name: **Gender**
- Right Click on the **students** resource on the canvas and then click **New Mapping**



- Name the mapping to the target class **Gender**



- **Reject** the suggested **Primary Identifier** and **Label**



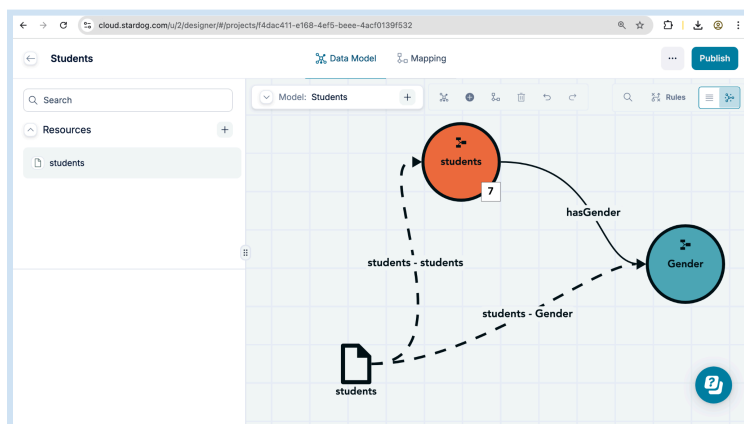
- Change both the **Primary Identifier** and **Label** to **Gender** and **Accept Changes**

Exercise 5 - Link Student Instances to a Gender Class

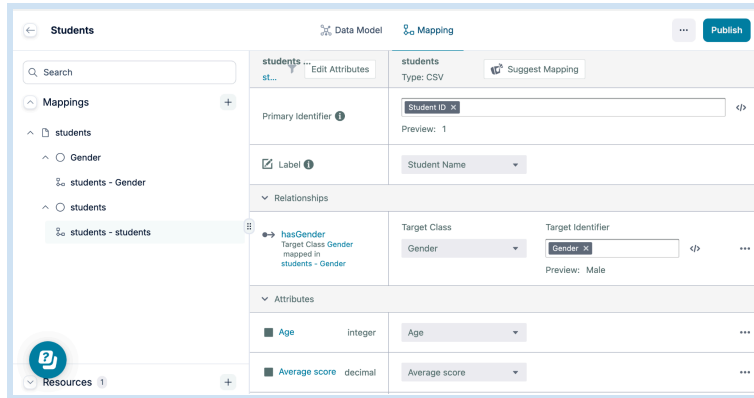
In this exercise you will learn how to link the student class instances to a gender class.

Step 1 - Create a relationship

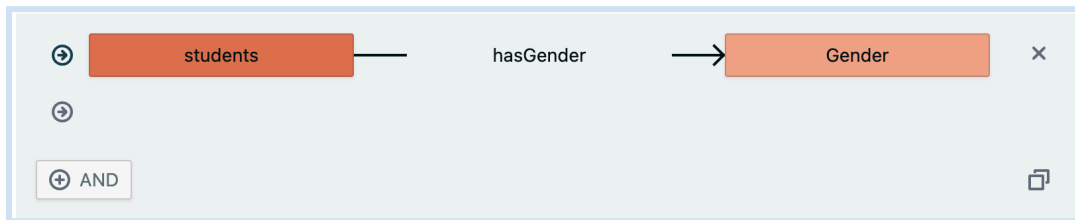
- Add a **Relationship**  and name it **hasGender** with a **Source** of **students** and a **Target** of **Gender**.



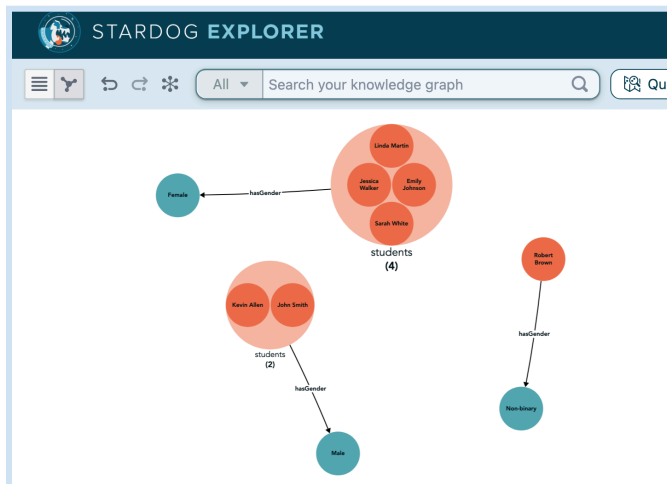
- Double click on the students-students mapping and open it up
- hasGender Target should be Gender and the Target Identifier should be Gender



- Publish your project using all the defaults from the last exercise
- Open Explorer and then open Query Builder
- Build this query



- Confirm you get results similar to these



Exercise 6 - Create and Map a New Class

In this exercise you will add and map a new class.

Step 1 - Add Remote as a Class

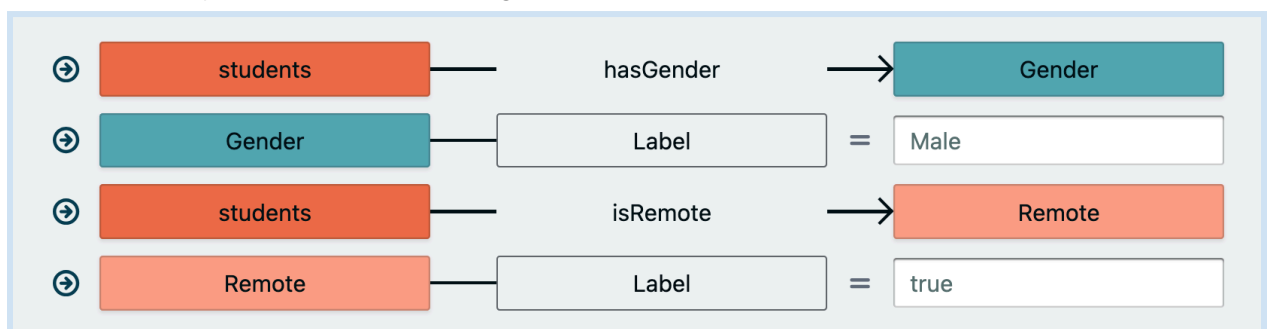
- Click on  to add a new class
- Class Name: **Remote**
- Right Click on the student resource on the canvas and choose **New Mapping**
- Choose a target class of **Remote**
- **Reject** the suggested **Primary Identifier** and **Label**
- Change both the **Primary Identifier** and **Label** to **Remote**

Exercise 7 - Link Student Instances to a Remote Class

In this exercise you will learn how to link the student class instances to a new Remote class.

Step 1 - Create a relationship

- Add a **Relationship** and name it **isRemote** with a **Source** of **students** and a **Target** of **Remote**.
- Double click on the students-students mapping and open it
- isRemote Target should be Remote and the Target Identifier should be Remote
- Publish
- Open Explorer and then open Query Builder
- Build this query to validate the changes to the data model

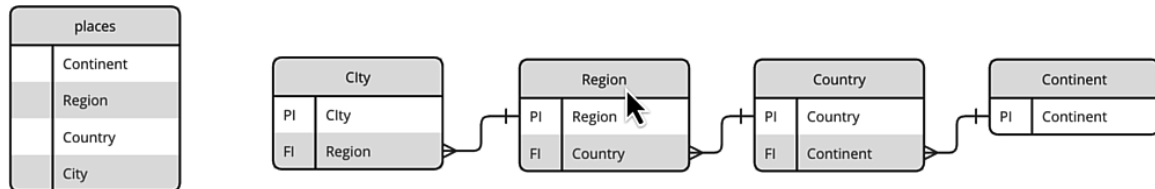


Confirm your query can identify the two male students who are remote, Kevin Allen and John Smith

Note: In production models, Gender and Remote might often remain just as attributes. Here we model them as classes to demonstrate instance linking.

Taxonomy Example

In this exercise, you will be building out a taxonomy using a .csv file containing geography data. Here is the entity relationship diagram (ERD) to use as a reference.



Exercise 1 - Create a Project and Upload Data

In this exercise, you create a new Designer project for the geography taxonomy and upload the data source.

Step 1 - Open Designer

- From <https://cloud.stardog.com/>, click on 'Stardog Designer'.

Step 2 - Create a new project

- From the Designer home screen, click on the 'New' button in the upper right corner.
- Give your project and Title and Description and choose **Create Project**

Exercise 2 - Map a .csv Resource

Step 1 - Download the .csv file

- [geography taxonomy.csv](#)
- Open the file and review the data. Think about what the primary key and label should be before you start.

Step 2 - Click the + next to resources to add a new project resource

- Click on **Choose files** and select the **geography taxonomy.csv** file

- Accept the defaults and click on Create & Map and Accept Changes. This will create a single class that will contain all our data. While this approach is not always required, it is generally good practice when working with flat files. Modeling individual aspects separately can sometimes result in certain attributes being omitted; data that you may later realise would have been useful to include.
- Accept all of the suggestions
- Once you see the class and the resource on the canvas, click on Mapping
- Click on the **geography taxonomy - geography taxonomy** mapping
- Edit the Mapping as shown

- Notice we're using a compound PI using all 4 attributes: Continent, Country, Region, City as shown in the image above. If we were to use a simpler Primary Identifier, such as City alone, different cities with the same name would generate identical IRIs. For example, both Santa Cruz in Bolivia and Santa Cruz in California would produce the IRI: **urn:geo:geography_taxonomy:Santa%20Cruz**

Santa Cruz	
City	
inRegion	California
	Sierra

To ensure each city has a unique IRI, we define a compound primary key that incorporates all four attributes. This results in distinct IRIs even for cities with the same name, such as:

urn:geo:geography_taxonomy:Europe__Bolivia__Sierra__Santa%20Cruz

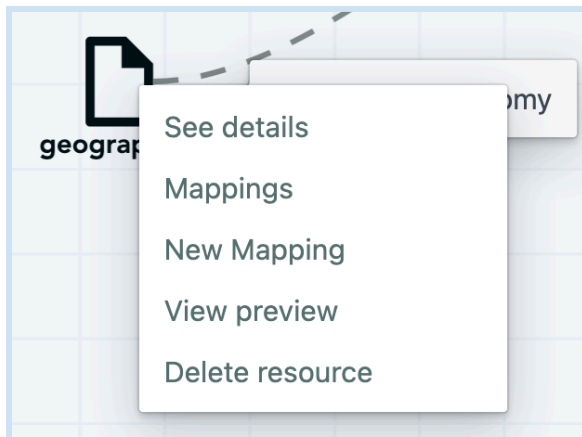
And

urn:geo:geography_taxonomy:North%20America__United%20States__California__Santa%20Cruz

- Make the **Label: City**

Step 3 - Extend the model with additional classes

- Create the following classes: **Continent, Country, Region and City**
- Right Click on the **geography_taxonomy** project resource on the canvas and choose New Mapping



- Create a mapping from the resource to each of the 4 new classes.
 - The compound PI for City should be Continent_Country_Region_City and Label City
 - The compound PI for Region should be Continent_Country_Region and Label Region
 - The compound PI for Country should be Continent_Country and Label Country
 - The PI for Continent should be Continent and Label Continent

Step 4 - Create Relationships

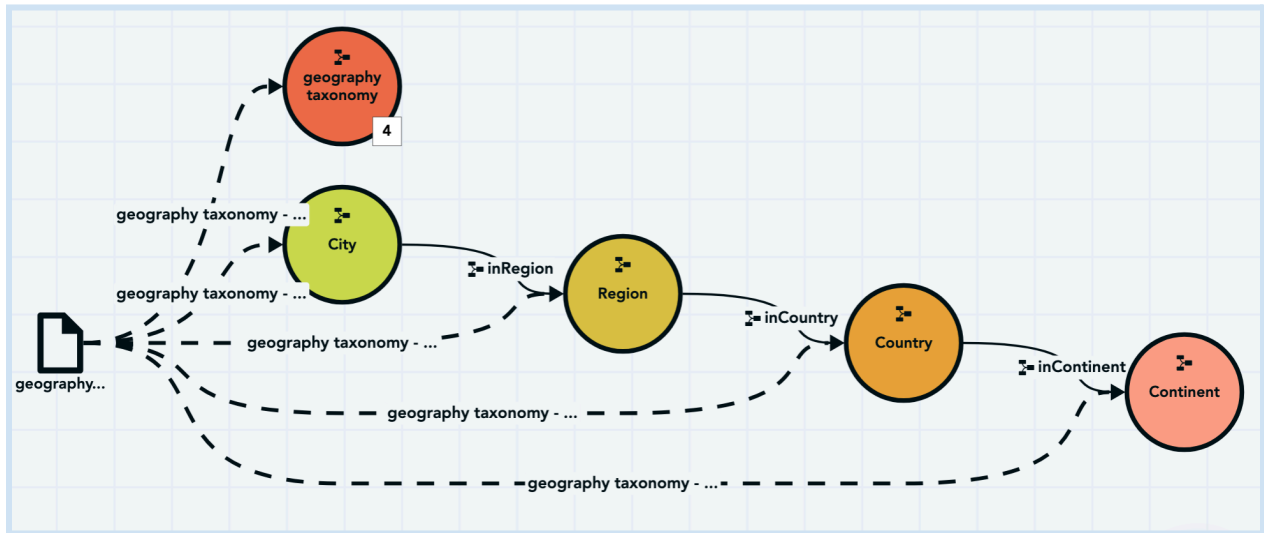


- Add a **Relationship** and name it **inRegion** with a **Source** of **City** and a **Target** of **Region**
- Add a **Relationship** and name it **inCountry** with a **Source** of **Region** and a **Target** of **Country**
- Add a **Relationship** and name it **inContinent** with a **Source** of **Country** and a **Target** of **Continent**

- Choose the redraw layout



- Your canvas should look something like this



Step 5 - Map Relationships using the target identifiers as shown

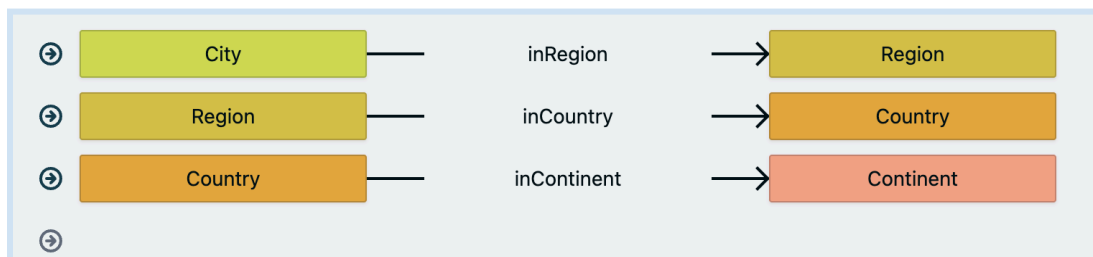
geograp...	geography taxonomy
City	Type: CSV
Primary Identifier	Continent x Country x Region x City x Preview: Africa_Egypt_Cairo_Cairo
Label	City
Relationships	
inRegion Target Class Region mapped in geography taxonomy - Region	Target Class: Region Target Identifier: Continent x Country x Region x Preview: Africa_Egypt_Cairo

geograp...	geography taxonomy
Region	Type: CSV
Primary Identifier	Continent x Country x Region x Preview: Africa_Egypt_Cairo
Label	Region
Relationships	
inCountry Target Class Country mapped in geography taxonomy - Country	Target Class: Country Target Identifier: Continent x Country x Preview: Africa_Egypt

geograp... Country	geography taxonomy Type: CSV	
Primary Identifier ⓘ	Continent X — Country X Preview: Africa__Egypt	
Label ⓘ	Country	
Relationships		
inContinent Target Class Continent mapped in geography taxonomy - Continent	Target Class Continent	Target Identifier Continent X Preview: Africa

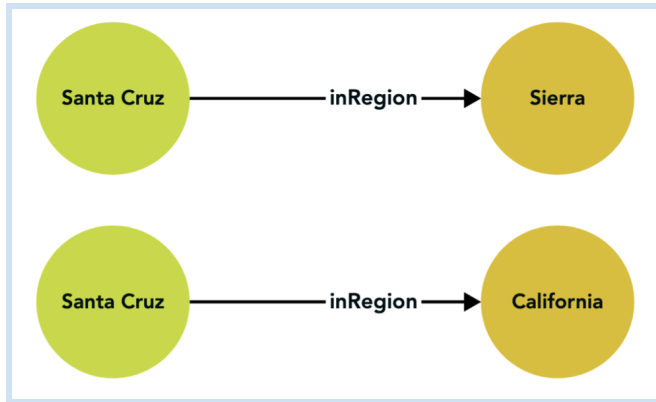
geograp... Continent	geography taxonomy Type: CSV	
Primary Identifier ⓘ	Continent X Preview: Africa	
Label ⓘ	Continent	

- Publish your new project and open in Explorer
- Create the following query using the **Query Builder**
- Save the query



- Go to **Options** and change the layout to **Vertical Tree**

- Confirm you have properly displayed your taxonomy. (Taxonomy usually means there is only one relationship eg inPlace rather than inRegion, inCountry etc. So our taxonomy is really a Hierarchy)
- Find Santa Cruz on the graph. Notice there are 2 and confirm they are distinct.



Many to Many Relationships

This exercise will show the steps to set up a data model that has a many to many relationship. In the example we will model Multiple Students taking the same course. And Each Student taking Multiple Courses.

Exercise 1 - Create a project and upload data

In this exercise, you create a new Designer project for **Courses** and upload the data source.

Step 1 - Open Designer

- From <https://cloud.stardog.com/>, click on `Stardog Designer`.

Step 2 - Create a new project

- From the Designer home screen, click on the `New` button in the upper
- Choose a **Title** and **Description** and **Create Project**

Exercise 2 - Map Project Resources

Step 1 - Download the .csv files

- [Courses.csv](#)
- [students.csv](#)

Step 2 - Add Project Resources

- Upload and auto-map students.csv and courses.csv
- Rename the classes eg Student and Course
- Modify attributes on the Student class; Average Score, Date and Remote to their correct data types as before.

- Choose appropriate Labels

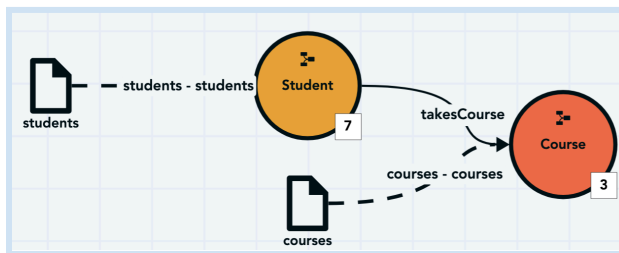
Exercise 2 - Create and Map Relationships

Step 1 - Download file

- [links.csv](#)

Step 2 - Add a Relationship

- Add a relationship called **takesCourse** from **students** and **course**



- Add a **Resource**
 - Upload the **links.csv** file
 - Accept the defaults and click Create (not Create and Map) . We need only the resource - not a class.

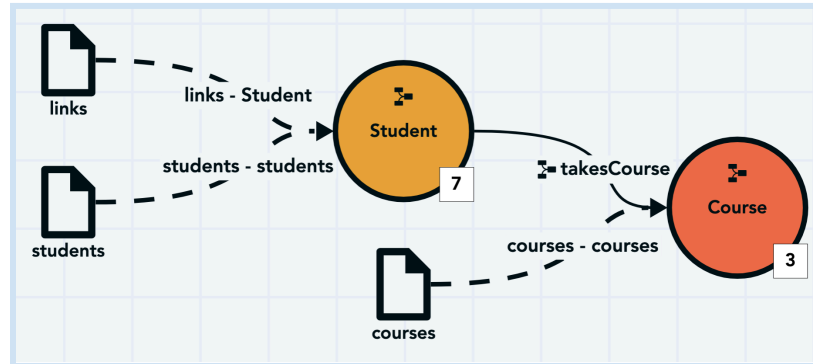
Step 3 - Edit the Course Class

- When reviewing the Links resource, you may notice that the CourseID column contains values such as **C-101**, whereas the Primary Identifier (PI) defined in the Course mapping is currently just **101**. To ensure these align correctly, we need to update the Primary Identifier for the Course class to prepend the prefix "**C-**". This will generate identifiers such as **C-101**, matching the foreign key value used when mapping the takesCourse relationship. If the identifiers do not match exactly, the relationship mapping will fail silently (no links created - ie no triples correctly connecting the students to their courses)
- Edit the **Course** PI so that it displays the **course ID** as C-101
 - Double click on **courses - courses** mapping
 - Add a C- as static text in front of the **CourseID** attribute

Step 4 - Map New Relationship

In this step you will map the **CourseID** and **StudentID** from the links resource to the new relationship called **takesCourse**.
 Right Click on the **links** resource and choose **New Mapping** to the **Student** class.

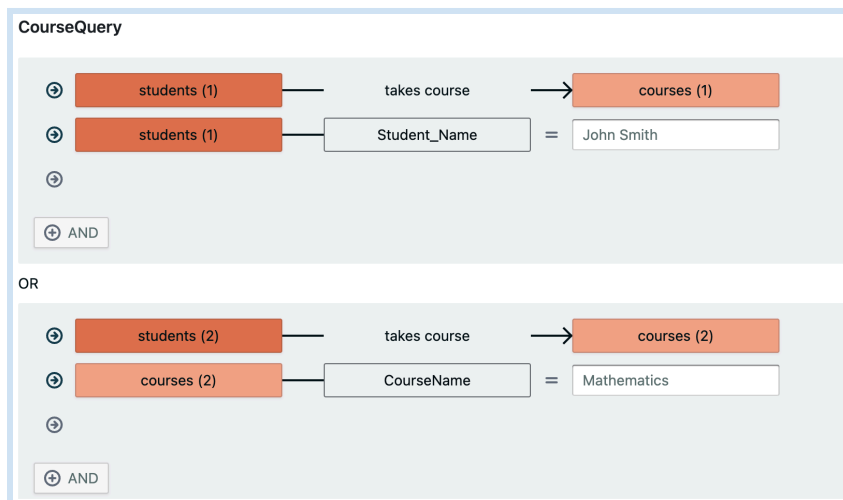
- Reject **Enrollment Date** as the PI and change it to **StudentID**. Don't assign a Label here. The label for Student comes from the mapping from the **students** resource.
- Go to the **takesCourse** relationship and make the **Target Class Course** and the **Target Identifier CourseID**. Your canvas should look similar to this



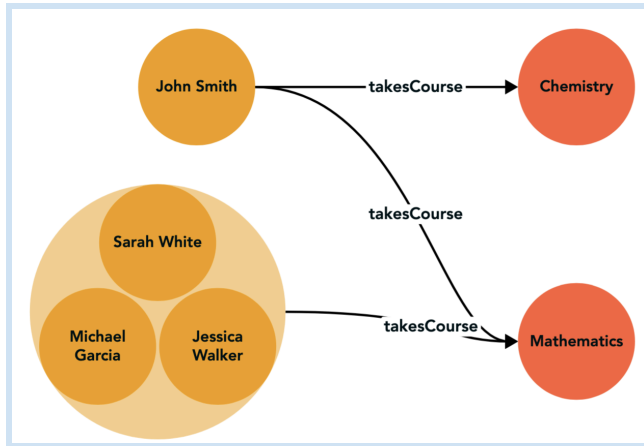
- Publish and open Explorer.

Step 5 - Validate in Explorer

This step provides an example of how to validate your data model using Explorer. \nOpen the Query Builder and build this query



- View and review the results.



The results show you have correctly mapped a many to many relationship. Multiple students take Mathematics and John Smith takes multiple courses.